GitHub Agentic Workflows

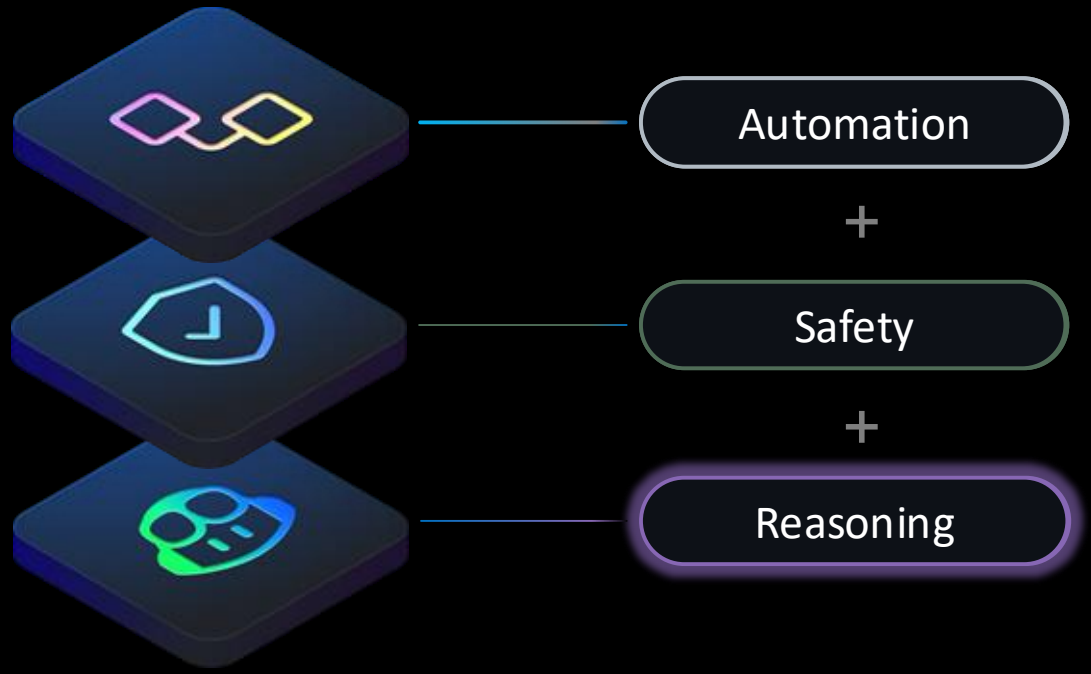
TECHNICAL PREVIEW

PELI DE HALLEUX
Principal RSDE
Microsoft Research

MADE WITH ♥
GitHub Next, Microsoft Research,
Azure Core Upstream & GitHub Actions

GitHub Agentic Workflows

Agentic human processes powered GitHub.



Workflows executed via GitHub Actions written in natural language.

Isolated sandboxes with scoped permissions and network control.

Agents interpret intent & operate on context across models. Copilot CLI · Claude · Codex · Gemini

Automate all the things

Agents accelerate the entire SDLC.

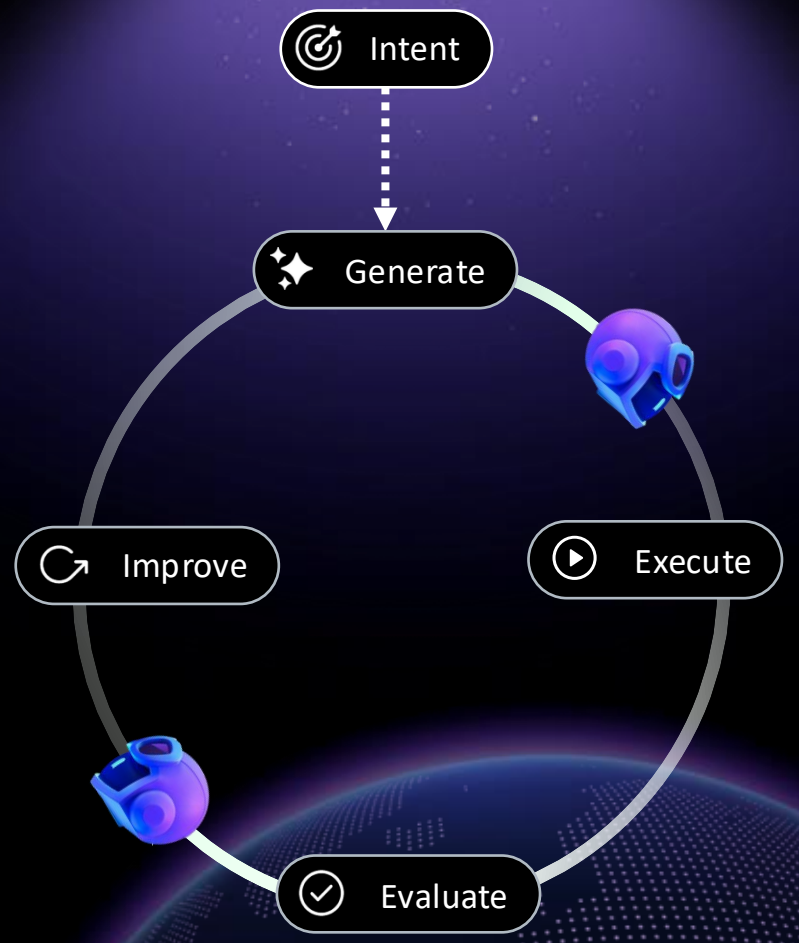
Every step of our software process can be “agentified”.

Automation is the true accelerator.

Completion: 1.4x, Dev solo: 10x, Automation: 100-1000x.

Scale with processes not swarms.

Orchestration not chaos.

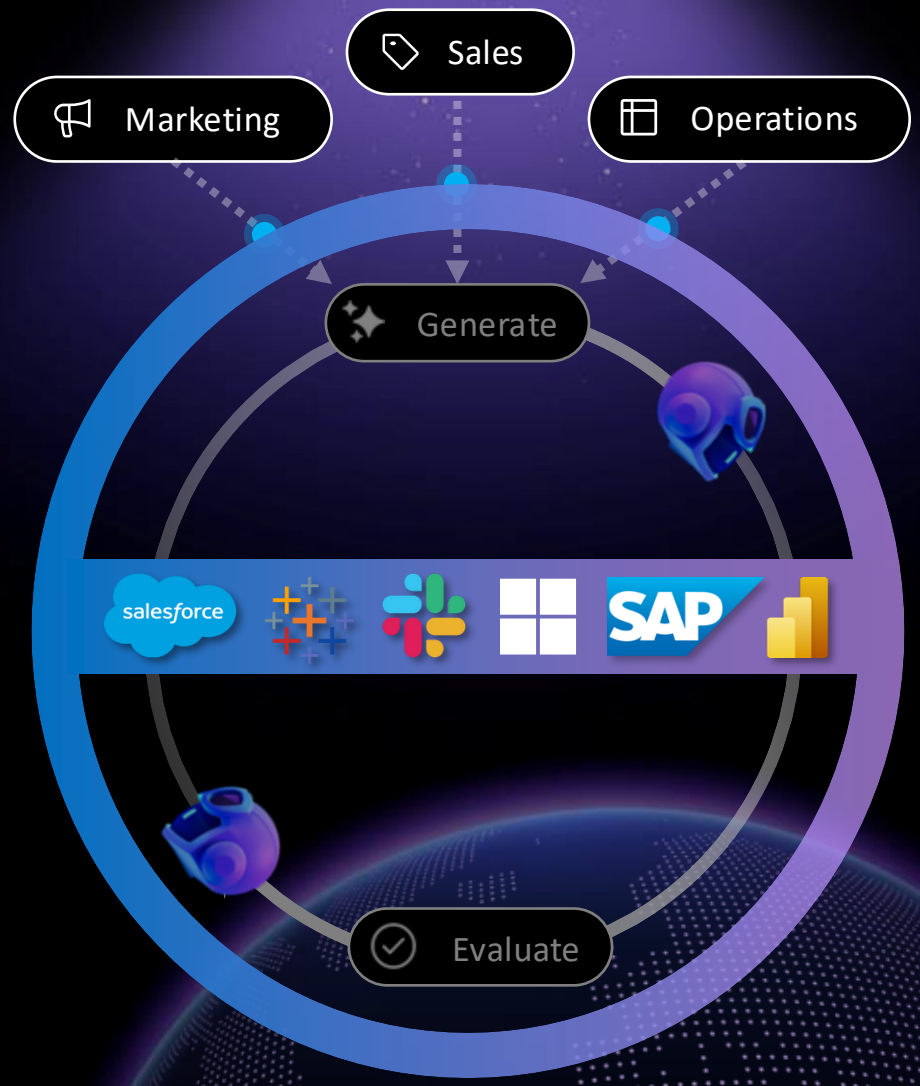


Automated Reasoning

for everyone & every organization

Agents can only reason about what’s accessible.

All business critical MCPs must be automatable!



Agents are dangerous

A sandbox enforces what the agent can access, do, and affect.

Prompt Injection

Permissions Abuse

Data Exfiltration

Malware Injection

Pull Requests
Issues
Web Pages

Inputs

Read-only Permissions
Zero Secrets
Containers/Firewalls
Integrity Filtering

Agent

Sandbox

Create Issue
Create Pull Request
Send Message

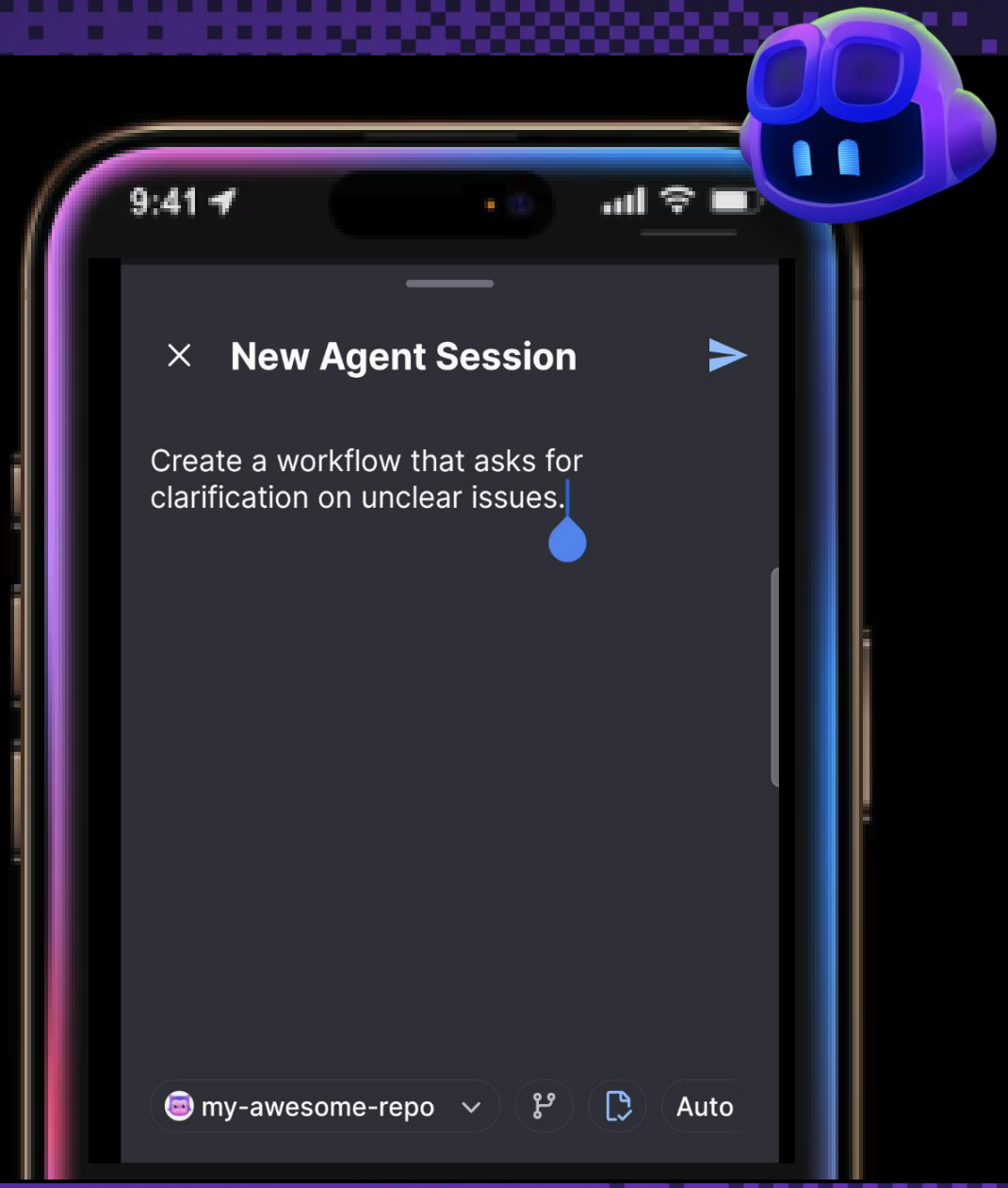
Safe Outputs

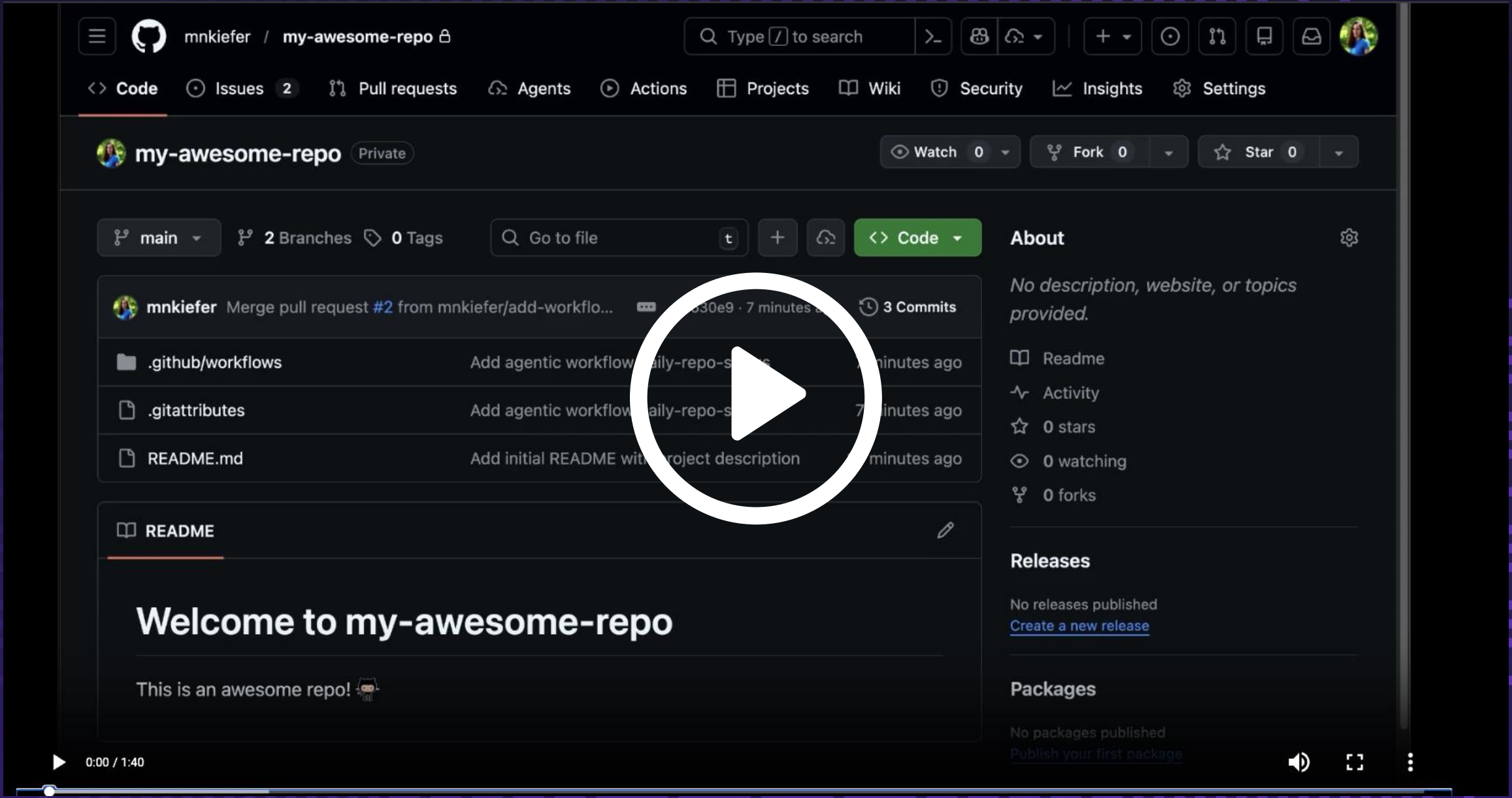
Intent-driven Automation
through Agentic Authoring

Write agents with the help of agents, iteratively.

- ✓ Configure your repository
  for GitHub Agentic Workflows
- ✓ Create new workflows
- ✓ Debug existing workflows
- ✓ Remix workflows between repositories

Work from anywhere




mnkiefer / my-awesome-repo
Type / to search
Code
Issues 2
Pull requests
Agents
Actions
Projects
Wiki
Security
Insights
Settings
my-awesome-repo
Private
Watch 0
Fork 0
Star 0
main
2 Branches
0 Tags
Go to file
t
Code
mnkiefer Merge pull request #2 from mnkiefer/add-workflo...
3 Commits
.github/workflows
.gitattributes
README.md
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
README
Welcome to my-awesome-repo
This is an awesome repo!
0:00 / 1:40

Orchestrated Automation

through Agentic Human Processes

issue-clarifier.md

.github > workflows > issue-clarifier.md > # Issue Clarifier

```
---
on:                          # Trigger: when to run
  issues:
    types: [opened]

permissions: read-all        # Security: read-only by default

safe-outputs:                # Allowed write operations
  add-comment:
---
# Issue Clarifier

Analyze the current issue and ask for additional details
if the issue is unclear.
```

Sandbox

Agent

mnkiefer opened 24 minutes ago

Please add better setup instructions.

Right now it's not clear how to get started, what prerequisites are needed, or what the expected workflow is.

Create sub-issue

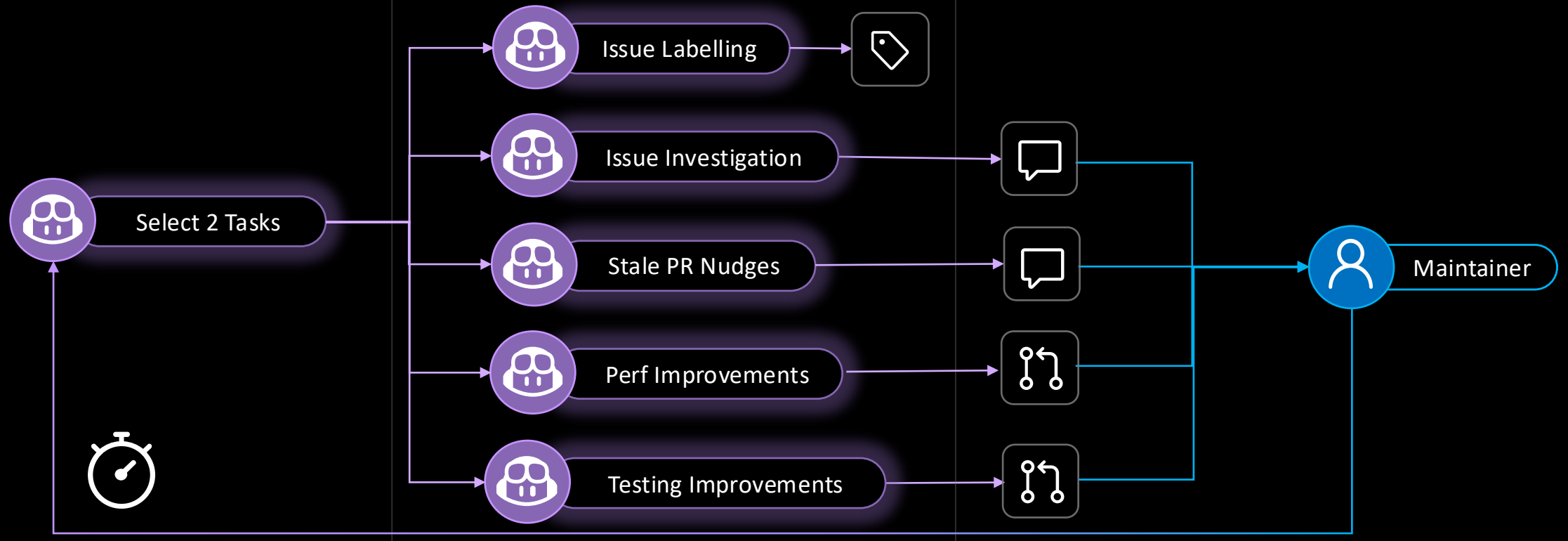
github-actions bot 20 minutes ago – with GitHub Actions

Thanks for raising this! The request for better setup instructions is clear, but to write something accurate and useful, a few more details would help:

1. **What does this project do?** A brief description of the project's purpose would help frame the setup instructions appropriately.

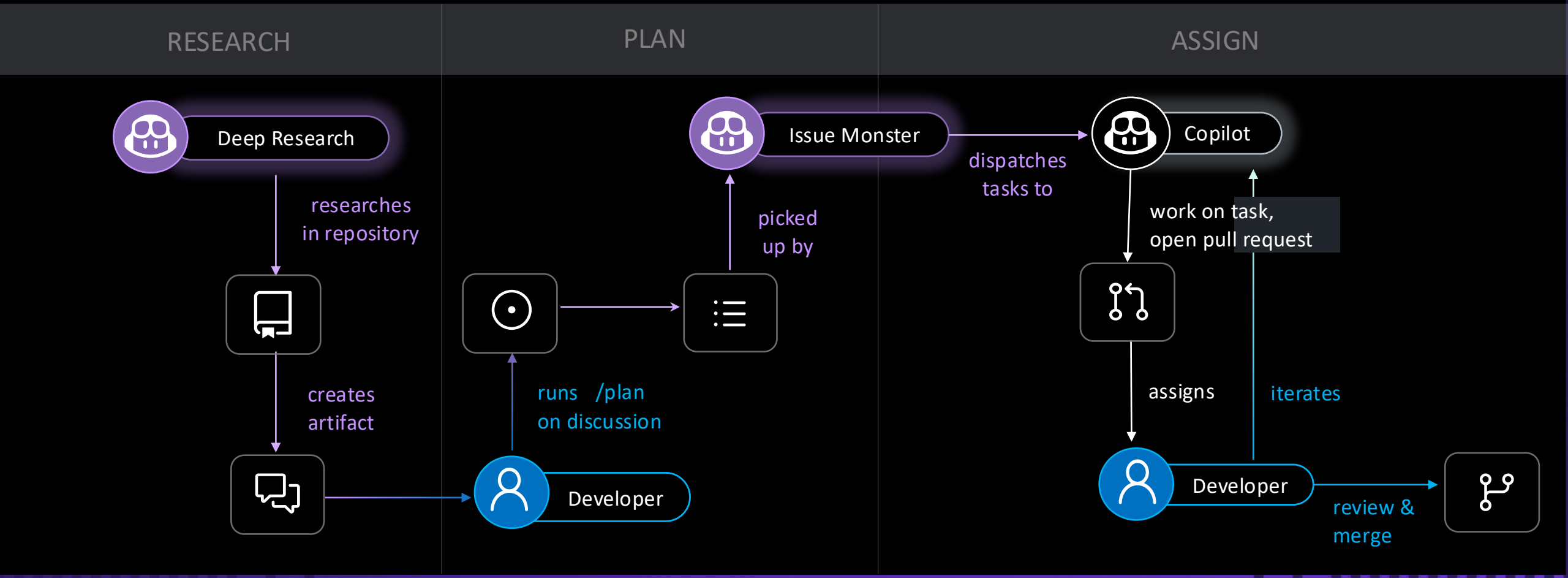
Repo Assist

Auto Maintenance Loop



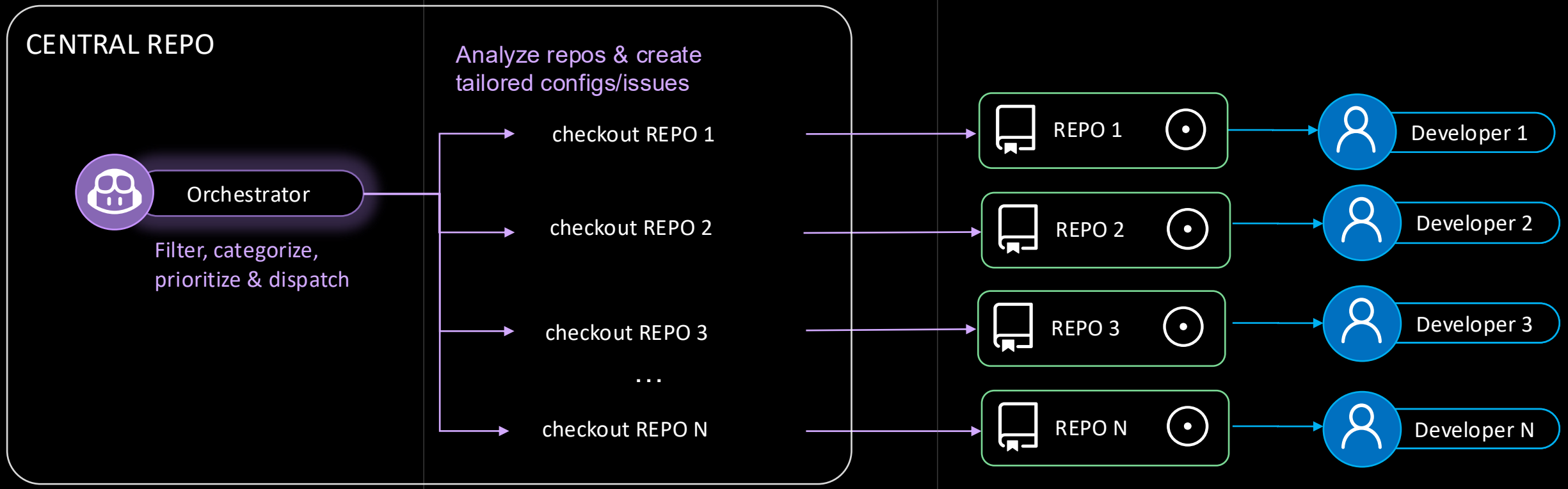
Research – Plan – Assign

Agentic Human Processes



Central Repo Ops

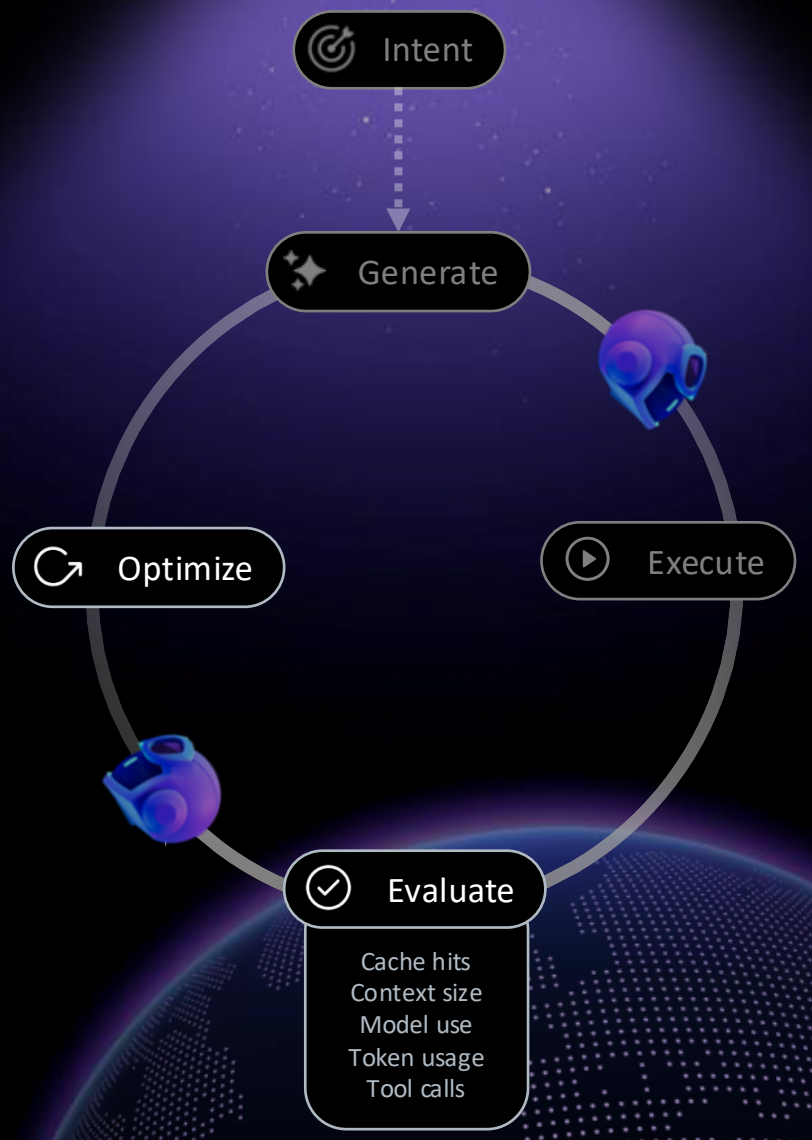
Rolling out large scale operations



Cost Self-Optimization

Agentic engineers must always optimize for:

- ✓ Fewer tokens
- ✓ Smaller models
- ✓ Agentic → Deterministic



Patterns

See our research for more useful design patterns.

[gh.io/gh-aw](gh.io/gh-aw)

Welcome to Peli's Agent Factory

Jan 12, 2026

Don Syme Peli de Halleux Mara Kiefer

Welcome, welcome, WELCOME to Peli's Agent Factory!

Imagine a software repository where AI agents work alongside your team - not replacing developers, but handling the repetitive, time-consuming tasks that slow down collaboration and forward progress.

Peli's Agent Factory is our exploration of what happens when you take the design philosophy of **"let's create a new automated agentic workflow for that"** as the answer to almost every opportunity that arises! What happens when you **max out on automated agentic workflows** - when you make and use dozens of specialized, automated AI agentic workflows and use them in practice.